

While it will be a while before we see Tesla cars on our roads, Qualcomm is using their platforms to help Indian car manufacturers to incorporate advanced features such as ADAS in their cars. We spoke with Uday Dodla, Director - Product Management at Qualcomm to learn more.

Q From an Indian market perspective, when do you see car manufacturers incorporating more of the experiences that are enabled by Qualcomm's Automotive and Ride platforms?

Uday Dodla: I'll start with an announcement that was made recently about the XUV 700. That would be the first car that has a truly digital cockpit that essentially combines the infotainment system – which has been there for a while, it is typically Android or Linux based – with the control cluster. The controls used to be mostly analogue and that has gone all digital. So all of this looks like a single panel. Now, this also has a few ADAS features such as automatic braking, lane change alerts and other basic stuff which is based on computer vision.

In India, we think that this is the first step towards going into advanced ADAS. With advanced ADAS, you need a lot more sensors such as LIDAR and at varying degrees things can get more complicated. However, we're already seeing companies such as Mahindra thinking about technology driving their portfolio. Since we deal with Indian OEMs quite extensively, I can tell you that there are plans that in the next 3-4 years, you will see more

and more ADAS making its way into the mid-tier level that are incorporating some key technologies of Qualcomm. The infotainment and cockpit technologies are a premium offering today but in a couple of years, everyone is going to have a digital cockpit with these becoming a standard offering.

And then, what do you do on top of a digital cockpit? You introduce ADAS features such as driver assist, driver monitoring, surround-view and more. These will all start making their way into vehicles in the not too distant future. You don't have to wait or buy a car from overseas, you will see these features in Indian cars very shortly.

Accelerating the Indian Automotive Ecosystem



Uday Dodla,
Director - Product Management
at Qualcomm

**We look at how
Qualcomm's platforms
are enabling Indian auto
manufacturers to bring
advanced features to
their cars**

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Q To what extent can car manufacturers add more features and take better advantage of the platforms? Also, do you believe that a custom OS or custom firmware ecosystem – like what we have for Android phones – emerge around these car platforms?

Uday Dodla: In the past, OEMs would look at one another and say, "Oh, they've got something, let us go and do that". That was good enough. Now, the trendsetters are clearly aiming for moonshots. Which is what made XUV do what they did. They've been at it for a couple of years. It's not easy to pull this off. There is a lot of work going into ensuring software stability since these vehicles are going at 100 km/h and you wouldn't want things going dark. There are road-worthiness tests and more.

What companies are aiming for and what Qualcomm is helping them achieve, is to provide enough horsepower that allows them to upgrade their systems in-market. You can roll out the vehicle with a certain set of features and over time you can perform software updates and introduce capability updates over the air because the hardware is already there.

In the past, the paradigm of 'cost' was overwhelming everybody. OEMs would just ask for a few features such as Bluetooth and call it a day. But now they're like, "We want to do a lot more, tell us how we can enable that?". So now, the technology and strategy teams in these OEMs are taking over. They're saying that they cannot be simply a manufacturing company, they have to be a technology company. They have to

be like a cell phone on wheels. They also have to connect with the customer once the vehicle leaves the showroom. Less than half the customers who buy a car actually go back to the dealership for servicing. With two-wheelers, that's close to 80 per cent.

Our platforms enable the OEMs to keep in touch with their customers, to understand their driving habits, their spending habits, enable preventive maintenance, etc. All these capabilities are being built-in. This means, that in a year or so, you will see more features being enabled in a vehicle such as the XUV700 that don't exist now. **Q**